

Assignment (Due date: 1st February, 2026) [Total Marks = 40]

1. Demonstrate with examples which functional group transformations are carried out by the following name reactions. State the reagents or catalysts (if any) used in these reactions. [2x3 = 6 Marks]
 - (a) Bayliss-Hillman reaction
 - (b) Wittig reaction
 - (c) Knoevenagel condensation
2. State two name reactions which can be used to convert phenol to salicylic acid. [2 Marks]
3. Is tetramethyl ammonium ion $[(CH_3)_4N^+]$ an electrophile or a nucleophile? Explain your answer. [2 Marks]
4. How many resonating structures are possible for p-nitrophenoxide ion? Draw them. [1+1 = 2 Marks]
5. Draw a mechanism for the following acid-base reactions. Label the acid, base, conjugate acid, and conjugate base. [2x2 = 4 Marks]
 - (a)



(b)



6. Draw the chemical structures of the possible tautomers of curcumin. Which isomer do you think will be more predominant in dichloromethane and why? [2+2 = 4 Marks]
7. Draw the E and Z isomers of the following: [4 x 2 = 10 Marks]
 - i) 4-Ethyl-3-methylhept-3-ene
 - ii) 3-Methyl-2-phenylhex-2-ene
 - iii) 1-bromo-2-methylbut-1-ene-1-thiol
 - iv) 2-bromo-1-iodoprop-1-ene
8. Calculate the energy difference (in kcal/mol) between the lowest energy and the next higher energy conformer of meso 2,3-dibromobutane, given the Gauche interactions (kcal/mol) for the following groups: Me-Me = 0.9; Me/Br = 0.25; Br/Br = 0.75 [2 Marks]
9. Locate the chiral center(s) in the following molecules (if any) and draw the R and S isomers. [4x 2 = 8 Marks]

